

**Table 3**

Average uncertainties of measurements and calculated quantities.

| Quantity |                           | Symbol           | Uncertainty        |
|----------|---------------------------|------------------|--------------------|
| Probes   | Temperature               | $u_c(T)$         | 0.1 K              |
|          | Pressure                  | $u_c(P)$         | 0.2%               |
|          | Water mass flow rate      | $u_c(\dot{m}_w)$ | 0.2%               |
|          | Surface roughness         | $u_c(R_a)$       | 0.17 $\mu\text{m}$ |
| Output   | Heat flux                 | $u_c(q)$         | 9.0%               |
|          | Heat transfer coefficient | $u_c(h_o)$       | 33%                |
|          | Falling film enhancement  | $u_c(K_{ff})$    | 5.2%               |

quantities were determined through the law of propagation of uncertainty. Equations were determined to calculate the uncertainties of these quantities based on the uncertainties of the probe measurements. The derivation of these equations was documented in detail by Christians [22] and the resulting calculated uncertainties are listed in Table 3. The uncertainties of the thermo-physical properties calculated with the aid of REFPROP 8 software [25] are listed in Christians [22].

The calculation of the uncertainty of the falling film enhancement factor  $K_{ff}$  in this study however differs from Christians [22]. Considering that  $K_{ff}$  is the ratio  $h_{o,ff}/h_{o,pb}$ , which are measurements of the same quantity under different conditions, the two input quantities are thus dependant. The subsequently derived uncertainty equation for  $K_{ff}$  was

$$u_c^2(K_{ff}) = K_{ff}^2 \left( \frac{u_c(h_{o,ff})}{h_{o,pb}} \right)^2 (K_{ff} - 1)^2 \quad (13)$$

The resulting average uncertainties as function of heat flux across the range of tests are shown in Table 3. These uncertainties are illustrated in Fig. 8 as function of heat flux for a typical case. The average HTC uncertainty was 33% (it varied between a minimum of 25% at a heat flux of 90 kW/m<sup>2</sup> and a maximum of 45% at a heat flux of 20 kW/m<sup>2</sup>). These uncertainties are comparable to previous researchers who worked on this same piece of equipment [7,31] and other similar experimental set-ups [10]. Measurements at heat fluxes lower than 20 kW/m<sup>2</sup> were not practical and were avoided due to the continued rise in the uncertainty of the HTC as the heat flux lowered. However, the average falling film enhancement uncertainty was significantly lower at 5.2% as it was calculated from two dependant properties. It varies from 8.7% at a heat flux of 20 kW/m<sup>2</sup> to 0.2% at a heat flux of 90 kW/m<sup>2</sup>.

## 5. Validation

The Wilson plots conducted acted as an initial validation stage, as the internal heat transfer modifying coefficient  $C_i$  (Eq. (4)) determined in this paper of 1.25 compared satisfactorily to previous researchers who worked with plain tubes on this same experimental setup with values of 1.27 [20], 1.18 [21] and 1.29 [34].

Condensation tests were used to validate the measurement system of the experimental apparatus as the literature for pool boiling and falling film studies often exhibit a wide spread of results as well as a lack of surface roughness information. Furthermore the most successful modelling solutions for boiling are semi-empirical as the production of correlations for nucleate boiling is fraught with difficulty [35], which makes them unsuitable for stand-alone validation purposes.

Condensation tests were chosen specifically as the Nusselt solution for condensation on tubes without inundation [3] is a solution that has been shown by previous work to agree well with experimental data, with previous results lying within 5% of the solution [36]. Thus the Nusselt solution could serve as a reference for comparison and validation. The results for a single horizontal plain copper tube with condensation on the outside of the tube at a saturation temperature of 30 °C with a surface roughness,  $R_a$ , of 0.15  $\mu\text{m}$  and no inundation are given in